

Esam Alselwi

Data Analyst | AI Specialist
Yemen, Sana'a

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Education

National University

Bachelor of Information Technology

October 2023 –Present (Expected graduation: February2027)

Yemen , Sana'a

Technical Skills

Languages : Python , Java , C++ , C# , PHP , HTML/ CSS , JavaScript , SQL
Databases: MySQL, PostgreSQL, SQLServer, Oracle, PL/SQL
Data Analysis & Visualization: Python(Pandas, NumPy, Matplotlib),Excel(Pivot Tables, VLOOKUP/XLOOKUP),Power BI(DAX, Dashboards),EDA
AI & Machine Learning: Python (Scikit-learn), LLMs, Prompt Engineering, AI Agents, API Integration, RAG, MCP
Web Technologies: ReactJS, NodeJS, Laravel
Cloud Developer Tools: AWS, Azure
Version Control: Git, GitHub
Technologies/Frameworks: Docker, GitHub Actions, Linux, CI/CD Pipelines, MLOps
Networking: Router/Switch Configuration, Routing Protocols (RIP, OSPF, BGP), VLANs, ACLs
Design Tools: Figma, Photoshop, Illustrator, Adobe Premiere Pro

Practical Experiences

- Applied data analysis techniques including data cleaning, exploratory data analysis, and visualization using Python, SQL, and Excel across multiple self-driven and academic projects.
- Designed interactive dashboards in Power BI to analyze trends, extract insights, and support decision-making.
- Built data-driven solutions using APIs and real-world datasets from multiple sources.
- Developed practical systems such as a **Pharmacy Management System** and an **E-Commerce Platform** using Java, JavaFX, PHP, and MySQL.
- Strengthened hands-on experience through university projects in databases, object-oriented programming, networking, and systems analysis and design.
- Gained practical experience in database integration, UI-to-database connectivity, and CRUD implementation across multiple projects.
- Improved the ability to transform requirements into clear, structured, and usable technical solutions with a focus on implementation quality.

Projects

Consumer Complaint Intelligence Platform | NLP, EDA, Data Visualization, Python, SQL

- **Objective:** To build an end-to-end analytics platform for processing and analyzing real-world financial complaint data from the CFPB.
- **Achievements:** Integrated demographic data with NLP to identify high-risk financial products. Developed interactive dashboards that pinpointed compliance risks, enabling data-driven risk management.

Crypto-Sentiment Correlation Engine | Time-Series Analysis, Sentiment Analysis, Plotly, Financial Data

- **Objective:** To investigate the dynamic relationship between cryptocurrency market fluctuations and macroeconomic sentiment.
- **Achievements:** Correlated Bitcoin price data with traditional market indices using time-series models. Quantified the impact of social sentiment on market volatility for predictive insights.

Emerging Tech Momentum Radar | GitHub API, Google Trends, K-Means Clustering, Forecasting

- **Objective:** To identify and track the momentum of emerging technologies by correlating search trends with open-source activity.
- **Achievements:** Leveraged GitHub and Google Trends APIs to build a data-driven intelligence tool. Applied K-means clustering to categorize tech adoption phases and technological shifts.

Global Talent Mobility Index | Python, Geospatial Analysis, World Bank API, Weighted Indexing

- **Objective:** To quantify global talent attractiveness using multi-dimensional indicators from the World Bank.
- **Achievements:** Engineered a production-grade data science engine to calculate a weighted talent index. Created geospatial visualizations to map global talent flows for strategic HR planning.